3.6 I/O CARTRIDGE BAORDS

3.6.1 VIEW DATA INTERFACE BOARD (FIGURES 4.19/4.20)

This board offers and interface to an asynchronous modem operating at 1200 B/S. The facility to send data from the P2000 via a backward channel at 75 B/S offers the possibility for a switched connection to the VIEW DATA systems. The main logic circuit on the PCB is the 8251 USART (IC11).
This programmable circuit interfaces between the systems databus and the serial V24 modem interface.

The USART is selected by the P2000 - CPU via decoder circuit (IC10) which is decoding addresslines A4 - A7 during an I/O instruction. The USART contains several registers, which are selected with this decoder and the command/data input, (wired to addressline A0).
This results in the next table for access and operation of the USART:

OUT 41	(after RESET) LOAD D0 -	D7 INTO MODE REGISTER (used to select char. format, parity etc)
OUT 41	(other times) LOAD D0 -	D7 INTO COMMAND REGISTER (used to activate transmit/receive actions)
IN 41		READ USART STATUS REGISTER (used to check on correct receive/transmit)
OUT 40		LOAD DATA INTO TRANSMIT REGISTER (character to be transmitted)
IN 40		READ DATA FROM RECEIVE REGISTER (fetch received character)

Circuits IC2 and IC3 are used to convert the modem interface levels (+12V/-12V) to TTL-level. The USART requires a transmission clock of 16x the actual bit-rate which is achieved by dividing the system clock Ø (2.5 MHZ) by 130 with help of IC 4,5 resulting in clock frequence of 16X 1200 BD. The circuit IC6 gives another 16 x division, thus offering the required clock of 16x75 HZ for the backward channel transfers. Selection of either normal transfer (2x1200 B/S) or backward channel transmission (viewdata) is software controlled via IC 8.

This two-bit register is loaded from the databus with an OUT 60H instruction, the address bus again being decoded via IC10. Bit 6 active selects multiplexer IC9 to offer the reduced transmit clock to the USART while modem signal CT106(CTS) is ignored, and forced active. Bit 7 is used to activate the backward channel modulator (RTS2, PIN19) before the characters are sent.

3.6.2 SERIAL INTERFACE BOARD (FIGURES 4.21/4.22)

The serial interface board can be used as a general purpose communication interface. This board offers some other features then the viewdata interface board described above. The USART 8251 (IC2) is a progammable controller for both asynchronous and synchronous control. It controls, apart from serializing an desserializing of data, most of the V24 modem control signals such as DTR, DSR, RTS and CTS.
Via a separate input port (IC3) the software can check the presence of some other control signals (DCD (8) and CIN (22).

The USART can be clocked either internally (strap 8,9) or externally (strap 7,10), however the external timing signals are not wired to the modem connector (TxC, pin 15 and RxC, pin 17).

The internal timing is derived from the systems clock Ø via a divider circuitry (IC 8,9). Switches S1 are used to select the required speed, closing one of the switches selects one timing out of a range of possible speeds.
The port selection on the board is performed by a binary to decimal decoder of address lines A4 - A7 (IC7). A decoding of binary 4 on these addresslines selects the USART.

On the USART addressline A0 is wired to the C/D input which results in the same table as for the viewdata board:

OUT 40	WRITE DATA TO LINE
IN 40	READ DATA FROM USART
OUT 41	SET USART MODE (AFTER RESET) SET USART COMMAND (NEXT TIME)
IN 41	READ USART STATUS

The decoder output 6 enables the other ports on the board, together with a decoding ot A0 and A1 this results in:

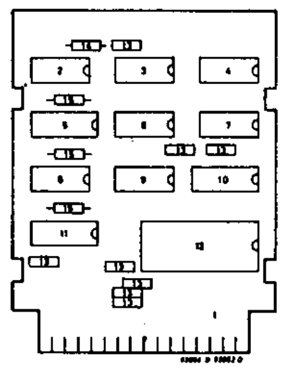
IN 61	READ STATUS PORT (DCD, CIN)
IN 62	READ GENERAL PURPOSE STRAPS

The last port is used to read a strap bank S2 via which the system can be informed about things like transmission type, required parity etc.
This strapping depends of the application and are accessable by the user.

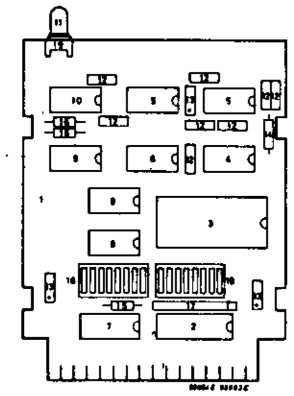
3.7 SIGNAL NAMES

A0 -15	16 bit address bus
AC0-11	Screen Refresh Counter (0-1919)
ACCL	Address Counter Clock
ACRES	Address Counter Reset
AE0-7	Address Bus from CPU board (T-version) or Monitor board (M-version)
AM0-6	Memory Address Bus
AM0-7	Multiplexed Address Bus
AU0-14	Address Bus to Upper boards
AV0-10	Address Bus to Video Memory
BEES	Beeper Select
BEEP	Activate the audible alarm
BET	Begin or End of tape
BS0	Bank Select 0 (A000H-DFFH)
BS1	Bank Select 1 (E000H-FFFH)
BS10	Bank Select 0 and 1
C95	Character 95 (last character)
CA0-6	Character Address
CARS 1	Cartridge selection 1 (1000H-2FFFH)
CARS 2	Cartridge selection 2 (3000H-4FFFH)
CAS	Column Address Strobe
CAS 0	Column Address selection bank 0
CAS	Column Address selection bank 1
CC0-6	Binary Court - Character Counter (- 96)
CIP	Cassette in position
C0585	Character 5 up to 85
C0686	Character 6 up to 86
CR0-6	Video Memory Data
CS 1, 2	Chip Select 1 and 2 - Video Memory
CS24, 13	Chip Select 3 and 4 - Video Attribute Memory
CTC SEL	Select Z80 CTC (Port 88-8B) timer
D0-7	8 bit bi-directional data bus
DBENA	Data Bus Enable Attribute Memory
DBENV	Data Bus Enable Video Memory
DBRDU	Data Rus Read Enable Attribute Memory
DBLI	Blinking Attribute
DCUR	Cursor Attribute
DC0-2	Binary Count - DOT Counter
DE0-7	Data Bus on Extension Board
DBENA	Data Bus Enable Attribute Memory
DBENV	Data Bus Enable Video Memory
DBRDU	Data Bus Read Enable Attribute Memory
DBLI	Blinking Attribute
DCUR	Cursor Attribute
DC0-2	Binary Count - DOT counter
DE0-7	Data Bus on Extension Baord
DEW	Data Entry window. Active during scan line 5 to 22 (20msec)
DGRA	Graphic Attribute
DINV	Inverse Attribute
DIRECTION	Define Head Direction
DISA	Disable Access
DISPS	Disable Refresh Selection (M-version)
DM0-7	Memory Data Bus, bi-directional

DOT 0	Dot 0 (first dot)
DOT 7	Dot 7 (last dot)
DUL	Underline Attribute
DU0-7	Data Bus to upper boards
DV0-6	Data Character to Row Memory
FCDK	Floppy Controller Data Acknowledge
FCDR	Floppy Controller Data Request
FCINT	Floppy Controller Interrupt
FCRD	Floppy Controller Read Control
FCTC	Floppy Controller Terminal Count
FCWR	Floppy Controller Write Control
FDCS	Floppy Disk Controller Chip Select
FDCSEL	Select UPD 765 FDC (Port 8C-8F) floppy controller
FNR	Floppy Not Ready
FRES	Floppy Controller Reset
ESTA	Floppy Status - Register Selection
FWD	Forward
GLR	General Line Reset
HOSY	Horizontal Synchronization
INDEX	Index pulse detection
INPSEL	Select Input Port (Port 90)
INS	Input Selection (I/O decoder)
INSEL	Input Selection (floppy disk controller)
INT	Interupt Request
IOE	Input/Output enable extension bard (Port 80-FFH)
IORQ	Input/Output Request
INVERT	Inverse Video Control
KBS	Keyboard Selection
KBIEN	Keyboard Interrupt Enable
LIN 11	Line 11 (last line)
LOCK	Disables Interrupt generation if extension board is used
L0-3	Binary Count - Line Counter
LOSE	Load Output Shift Register
MBEN	Memory Bus Enable
MBEN 2	
MOTON	Motor On
MRDE	Memory Bus Read Enable
MREQ	Memory Request
NEWR	New Row
NMI	Non Maskable Interrupt
Ø	2,5 MHz Clock
Ø2	5 MHz Clock
OUTSEL	Select Output Port (port 90)



Pos.	Code Number	Description
A		View data interface
1B	5103 108 01240	PCB view data interface compl.
2C	9332 873 50112	IC 74LS138B
3C	9334 282 20112	IC INS 8251
4C	9332 876 50112	IC N74LS290A
5C	9334 510 30112	IC N74LS161AN
6C	9332 876 60112	IC N74LS293A
7C	9332 874 00112	IC N74LS157B
8C	9334 617 20112	IC N74LS74AN
9C	9332 869 30112	IC N74LS04A
10C	9332 963 20112	IC MC1488
11C	9332 912 00112	IC MC1489
12C	9332 886 70112	IC N74LS02A
13C	2012 572 10011	Capacitor 100nF 63V
14C	2322 211 23181	Resistor 180E
15C	2322 211 23102	Resistor 1K



Pos.	Code Number	Description
A		Serial Interface
1B	5103 108 01480	PCB Serial Interface compl.
2C	9334 452 40112	IC DM81LS97
3C	9334 282 20682	IC INS8251
4C	9333 700 80112	IC 74LS126N
5C	9332 912 00112	IC MC1489
6C	9332 869 30112	IC N74LS04A
7C	9332 873 50112	IC N74LS138B
8C	9332 278 30112	IC 74LS393
9C	9334 870 70112	IC N74LS27A
10C	9332 963 20112	IC MC1488
11C	9332 184 80112	LED SR103D
12C	2222 640 03103	Capacitor 10nF 63V
13C	2013 017 01005	Capacitor 0,47µF 35V
14C	2322 211 23221	Resistor 220E
15C	2322 211 23122	Resistor 1K2
16C	2322 211 23332	Resistor 3K3
17C	3103 200 57740	Resistor network 4310R-101-472
18C	2422 126 01126	Dip switch
19C	8203 235 10740	LED socket